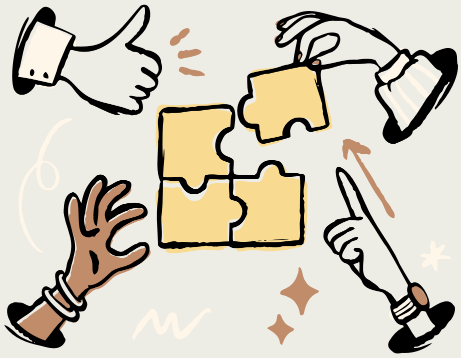


SHOTIFY.CO

Design documentation

Problem Statement:

People struggle to take passport-approved photos because of strict rules for pose, lighting, and background. This often leads to rejected photos, wasted time, and extra costs. An AI-guided mobile app is needed to help users take compliant passport photos easily and correctly on their own.

Project summery:

"Hi, I'm Krishaang and this is **Shotify**.

Music wouldn't be the same without **Spotify**, and e-shopping wouldn't exist without **Shopify**. Now, it's time for **Shotify** — an app that makes taking passport photos simple.

Using **AI and voice guidance**, Shotify tells you how to pose, fixes your background, and even takes the photo for you — no clicks needed. It checks everything with **official government standards**, so your photo's ready to go in seconds.

Shotify — the smart way to take your perfect shot."

Overview

Product	According to the survey the app had received a 4.5 out of 5 so this proves that the app is unique and will be used by many people.
Specifications	● People Said that the photo session was easy to run about 85% ● The layout did not have a sever lag ● The app was really beneficial and would be shared to others
Key features and benefits	● AI Guidance & Government Verification ● Easy to Use & Simple Navigation ● Convenience & Accessibility
Price point	Shotify is designed to be **easy, smart, and affordable** for everyone. The first **two photos are completely free**, so users can try the app with no cost. After that, it's just **$1 per month** for unlimited verified, high-quality passport photos — a fraction of what traditional photo studios charge.

Risk-mitigation:

-The users might not trust the use of government verify checkers.

Solution- The app is now not going to divert the user to some-other website it is going to have an inbuilt checker for the picture.

Customer feedback

Sources

List channels through which feedback is collected.

- Online surveys (pro's)
  - "Easy to use and simple to understand."
  - "Clean, sleek, and professional design."
  - "Very accurate and convenient — saves time and money."

- Online surveys (con's)
  - "Could use more features and customization."
  - "Some buttons and loading times need improvement."
  - "Would be better as a mobile app instead of a website."

Analysis

Interpret feedback as a team by identifying major themes and providing insights and interpretations.

Comment

- "Could use more features and customization."

Solution:

- A background option of your choice
- An auto navigating feature where you don't press any- buttons during your photo section

Comment

- "Some buttons and loading times need improvement."

Solution:

- We can add a speed monitoring system the makes changes according to the speed of the internet.
- I can add a reload button when a lag is occurred.

Comment

- "Would be better as a mobile app instead of a website."

Solution:

- We can make a app version for this website
- It will be avalable in both options.

Tech stack

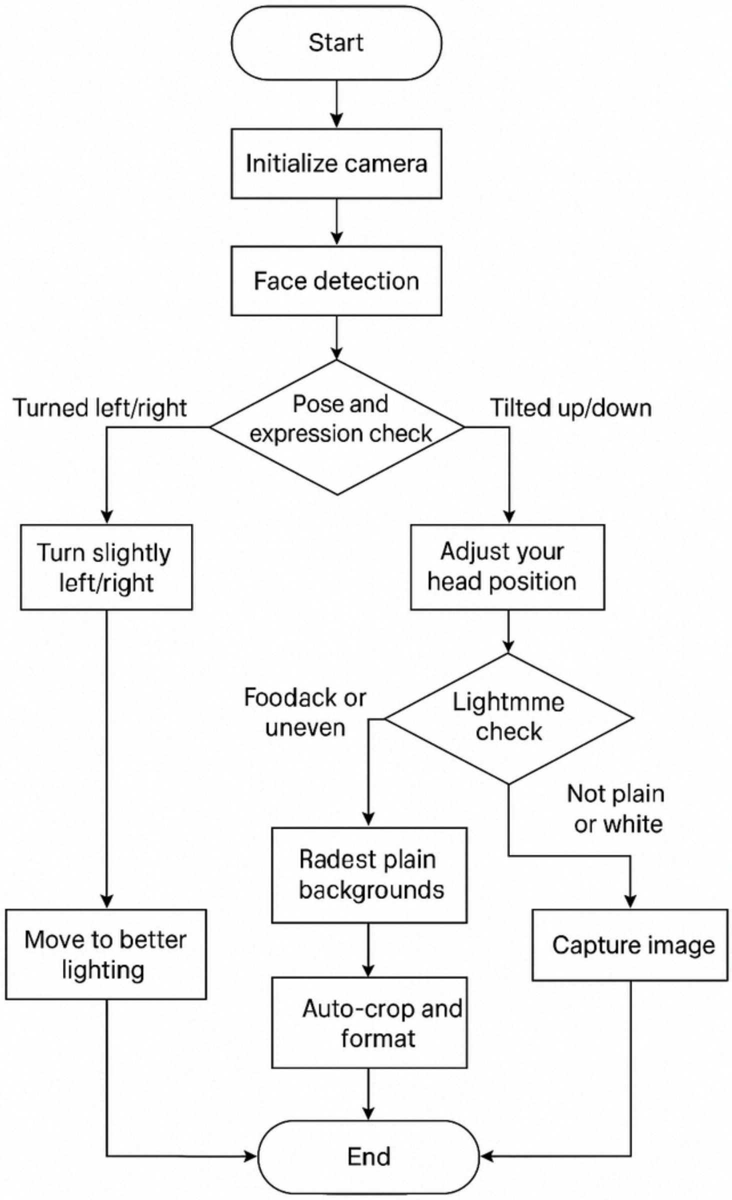
Tech Stack

Category	Tools / Technologies	Purpose
Frontend (Mobile App)	Flutter, React Native, Swift, Kotiin	Build UI and mobile app screens
Backend	Node.js (Express), Python (FastAPI/Flask), Firebase	Handle APIs, AI processing, and app logic
AI & Computer Vision	MediaPipe, OpenCV, TensorFlow Lite, PyTorch Mobile, U^2-Net/MODNet	Face detection, angie check, lighting/ background analysis, background removal
Cloud Services	Google Cloud Vision, AWS Rekognition, Firebase Storag, AWS S3	Cloud-based image validation and stora-ge
Authentication	Firebase Auth, Auth0, OAuth (Google/Apple)	Secure user login
Database		Store user data, photo history, settings
DevOps & Deployment	Docker, GitHub Actions, Firebase Hosting, Vercel	Deployment, CI/CD, automated builds

Algorithm

Action	explanation
Step 1: Start	Launch the app and open the “Start Photo Session” screen.
Step 2: Initialize Camera	Display a face outline or oval guide on screen.
Step 3: Face Detection	Locate the user’s face in real time. Detect key facial points (eyes, nose, mouth, chin).
Step 4: Pose and Expression Check	Calculate face angle using eye and nose coordinates.
Step 5: Lighting Check	If lighting is too dark or uneven, show “Move to better lighting.”
Step 6: Background Check	If background is not plain or white, show “Use a plain light background.”
Step 7: Capture Image	When all checks pass (pose, light, background, expression):
Step 8: Auto-Crop and Format	Crop image to standard passport dimensions (e.g., 2x2 inch ratio).
Step 9: End	Save or allow user to retake.

Flow-chart


Start
Initialize camera
Face detection
Pose and expression check
Turned left/right
Tilted up/down
Turn slightly left/right
Adjust your head position
Foodack or uneven
Lightmme check
Not plain or white
Radest plain backgrounds
Move to better lighting
Capture image
Auto-crop and format
End

Time-line

Month	Milestone	Description
Month 1 – Planning & Research	**Project Scope Defined**	Outline the app idea, define user needs, study passport photo standards, and decide core features (AI guidance, auto-capture, background removal).
Month 2 – Design	**UI/UX Wireframes Completed**	Create home screen, photo session screen, help/support screens, and interactive prototype for testing flow.
Month 3 – Core AI Module	**Face Detection & Angle Tracking Ready**	Integrate Mediapipe/FaceMesh for real-time head angle detection, facial landmark tracking, and pose feedback.
Month 4 – Auto-Capture & Voice Guidance	**AI Voice Assistant Implemented**	Develop text-to-speech guidance; set auto-capture triggers when face alignment and lighting are correct.
Month 5 – Background & Cropping	**Background Removal & Formatting**	Implement white background replacement, automatic cropping, resizing to passport standards, and preview/save feature.
Month 6 – App Integration	**Prototype Integration Complete**	Combine camera, AI guidance, auto-capture, background processing, and UI into a working prototype.
Month 7 – Testing & Debugging	**Internal QA & Bug Fixing**	Test all features, refine AI instructions, correct edge cases (bad lighting, different angles, multiple devices).
Month 8 – Beta Testing	**User Feedback Collected**	Share prototype with small group of users; gather feedback on usability, clarity, speed, and satisfaction.

Month 9 – Iteration	**Enhancements Implemented**	Improve UI/UX, optimize AI guidance, adjust thresholds for auto-capture, fix reported bugs from beta.
Month 10 – Launch Preparation	**Final App & Presentation Ready**	Polish prototype for final presentation or limited release, prepare documentation, marketing plan, and proposal visuals.